

# Mateusz Jaszcuk

☎ +1 (765) 746 9700 ✉ [jaszczukmatt@gmail.com](mailto:jaszczukmatt@gmail.com) 🎓 Scholar in Mateusz Jaszcuk 🌐 [mateuszjaszcuk.com](https://mateuszjaszcuk.com)

## Education

---

### University of Pennsylvania

*Ph.D in Mechanical Engineering and Applied Mechanics*

**GPA:** 4.00/4.00

*Philadelphia, PA*

*Aug 2026 – Present*

### University of Pennsylvania

*M.S.E. in Mechanical Engineering and Applied Mechanics – Mechatronics and Robotics*

**GPA:** 4.00/4.00

*Philadelphia, PA*

*Aug 2024 – May 2026*

**Thesis:** *Learning to Feel: Force-Aware Data-Driven Estimation and Control for Adaptive Physical Interactions*

### Purdue University

*B.S. in Aeronautical and Astronautical Engineering – Autonomy and Control*

**GPA:** 3.71/4.00

*West Lafayette, IN*

*Aug 2020 – May 2024*

## Skills

---

**Programming & Tools** - Python (PyTorch, JAX), C++, Isaac Lab, MuJoCo, MATLAB, ROS, PyBullet, Drake, CUDA

**Machine Learning** - Reinforcement Learning, Imitation Learning, Behavior Cloning, Diffusion, Sim-to-Real (Domain Randomization), Neural Networks (MLP, CNN, Transformers, NeRF), Probabilistic Modeling

**Control & Estimation** - Impedance Control, Operational Space Control, Inverse Dynamics, Optimal Control (MPC, iLQR, DP), State Estimation (EKF, UKF, PF), Safety-Critical Control (CLF-QP, CBF-QP), System Identification, SLAM

**Hardware** - Franka Research 3 (7-DoF), LiDAR, RealSense RGB-D, OptiTrack Motion Capture, Sensor Integration

## Experience

---

### Figuroa Robotics Lab (GRASP)

*Graduate Research Assistant*

*Jan 2025 – Present*

*Philadelphia, PA*

- Developed *Rapid Mismatch Estimation* – a real-time data-driven framework identifying unknown payload dynamics *online* on a 7 DoF manipulator, enabling safe and reactive manipulation under load uncertainty; published at CoRL 2025.
- Implemented an operational space impedance controller with a DS-based motion policy, enabling compliant and reactive manipulation; deployed a full pipeline on Franka Research 3 via custom ROS packages, validated across sim and hardware.
- Designed a transformer-based interaction-aware adaptive controller that fuses point cloud and force modalities via self-attention over stacked latent tokens, enabling real-time classification of human perturbations vs. payload loads – achieving simultaneous unknown load compensation, precise tracking performance, and compliant response to physical interactions.
- Supported research on learning DS from demonstrations, contributing to imitation learning-based policy development.

### Composite Manufacturing & Simulation Center

*Undergraduate Research Assistant*

*Oct 2022 – May 2024*

*West Lafayette, IN*

- Developed a Physics-Guided Transfer Learning framework using Bayesian model with Gaussian Process surrogates for data-efficient composite's property prediction across varying conditions, reducing required characterization efforts.
- Co-authored a peer-reviewed journal publication and designed visualizations to demonstrate the model's evaluation.

### Air Force Research Lab - Purdue UAS Research and Test Facility

*Undergraduate Research Assistant, Project Lead*

*Aug 2023 – Dec 2023*

*West Lafayette, IN*

- Developed autonomous fixed-wing aircraft for enclosed hangar-flight with Windracers (UK), including sizing, flight simulations, and led motion capture integration for indoor localization and autonomous safety protocols validation.
- Managed project coordination, aligning the development cycle with the client's requirements.

## Awards

---

### Outstanding Research Award

Awarded by Department of Mechanical Engineering, University of Pennsylvania

Philadelphia, PA

May 2026

## Teaching Experience

---

### University of Pennsylvania

MEAM 5200 Introduction to Robotics - *Teaching Assistant*

Philadelphia, PA

Fall 2025

## Professional Service

---

IEEE International Conference on Robotics and Automation (ICRA) – Reviewer

IEEE Transactions on Robotics (T-RO) – Reviewer

Conference on Robot Learning (CoRL) – Reviewer

## Project Experience

---

### Small-Scale Transformer for Text Generation - Graduate Course Project

Aug 2024 – Dec 2024

*Student Researcher, Developer*

Philadelphia, PA

- Implemented a lightweight GPT-inspired transformer neural network to generate coherent artificial text.
- Designed multi-head attention, positional encoding, and MLP modules in PyTorch, trained the model on small datasets (books, screenplays) to demonstrate artificial text generation and performance evaluation.
- Conducted ablation on regularization (dropout, layer normalization) to evaluate the impact on performance.

### Autonomous Mobile Battle Robot - Graduate Course Project

Aug 2024 – Dec 2024

*Robotics Engineer, Team Member*

Philadelphia, PA

- Designed a mobile robot for autonomous competition task execution (wall-following, navigation, attack) across a full project lifecycle: concept design, prototyping, integration, testing, and final competition.
- Integrated mechanical and electronic subsystems (platform, motors, ultrasonic/ToF sensors, ESP32 microcontroller, custom circuits) and programmed an Arduino-based embedded system for real-time control.

### NOAA Wind Tracking Satellite Constellation - Senior Design Project

Aug 2023 – Dec 2023

*Attitude Dynamics and Control Engineer, Team Member*

West Lafayette, IN

- Developed a MATLAB 6-DOF model to simulate satellite attitude dynamics across multiple orbital planes.
- Coordinated system integration and designed active/passive attitude control strategies, ensuring the spacecraft met stability and maneuverability requirements for long-duration Earth's surface monitoring.

### Purdue Space Program NASA Student Launch

Aug 2021 – May 2023

*Project Manager (2022-2023), Launch Vehicle Construction Engineer (2021-2022)*

West Lafayette, IN

- Led a 60-member interdisciplinary team through the full project life cycle for the NASA Student Launch competition.
- Coordinated project deliverables and communications with NASA, ensuring the system meets mission requirements.
- Led launch vehicle development and 6-DOF flight simulations for trajectory optimization and mission design.

## Publications and Presentations

---

- [1] **M. Jaszczuk** and N. Figueroa, "Rapid mismatch estimation via neural network informed variational inference," in *9th Annual Conference on Robot Learning (CoRL)*, 2025.
- [2] A. J. Thomas, **M. Jaszczuk**, E. Barocio, G. Ghosh, I. Bilonis, and R. B. Pipes, "Probabilistic physics-guided transfer learning for material property prediction in extrusion deposition additive manufacturing," *Computer Methods in Applied Mechanics and Engineering*, vol. 419, p. 116660, 2024.
- [3] A. J. Thomas, **M. Jaszczuk**, E. Barocio, I. Bilonis, and R. B. Pipes, "Physics-guided transfer learning for property prediction in composite additive manufacturing." SAMPE Japan, 2023.
- [4] **M. Jaszczuk**, "A bayesian framework for transfer of process-dependent material properties." Purdue Fall Undergraduate Research Expo, November 2023. West Lafayette, IN, USA.